Warn clients two hours before a flight goes wrong, and turn a bad surprise into a managed one

1 in 4
flights from New York were late or cancelled from January to September 2023 (25.7% of 330,883 flights) [01]

2 hours
is the notice the agency can give; today clients get none

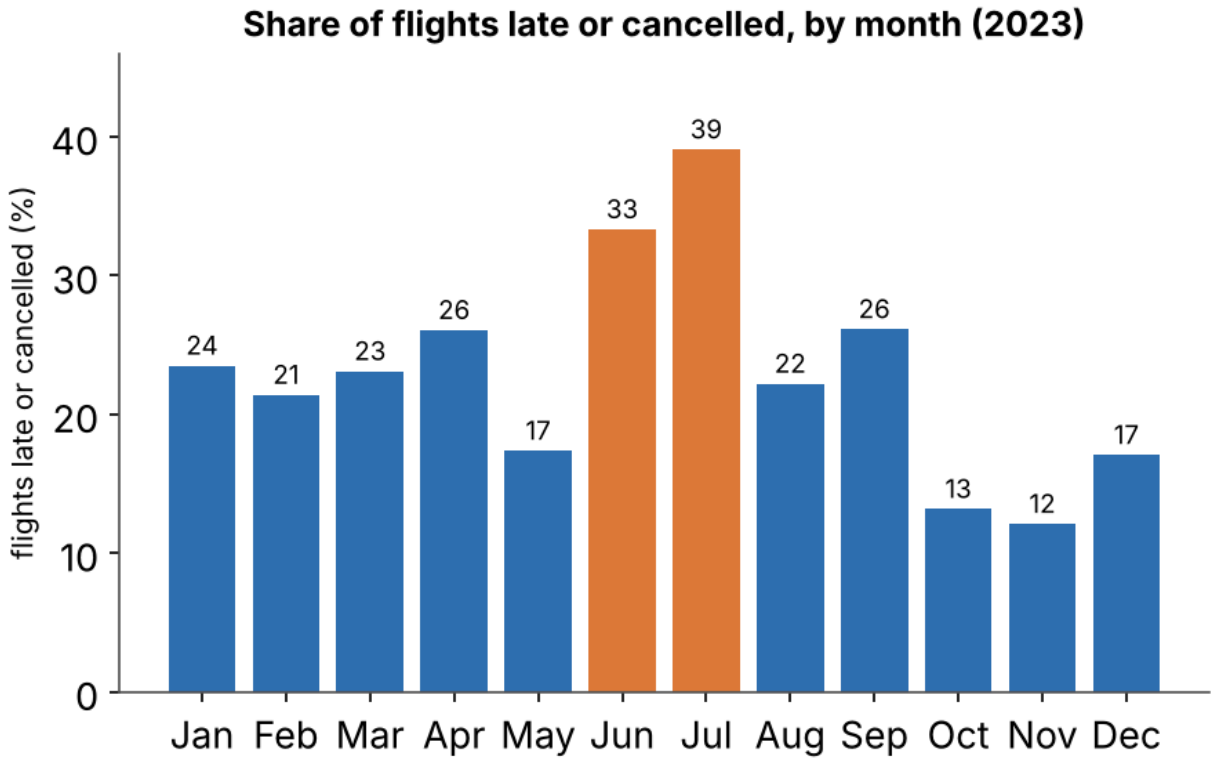
2 alerts
we can switch on now, tested on three months the model never saw [07]

Proposal for the agency owner: **approve two warning alerts and a weekly check.**

Data: nycflights23, every flight from JFK, LaGuardia and Newark in 2023 (435,352 flights). Mirna Embaby, Queen's University.

Sources: notebooks 01, 07

The problem is big, uneven, and clients hear about it too late



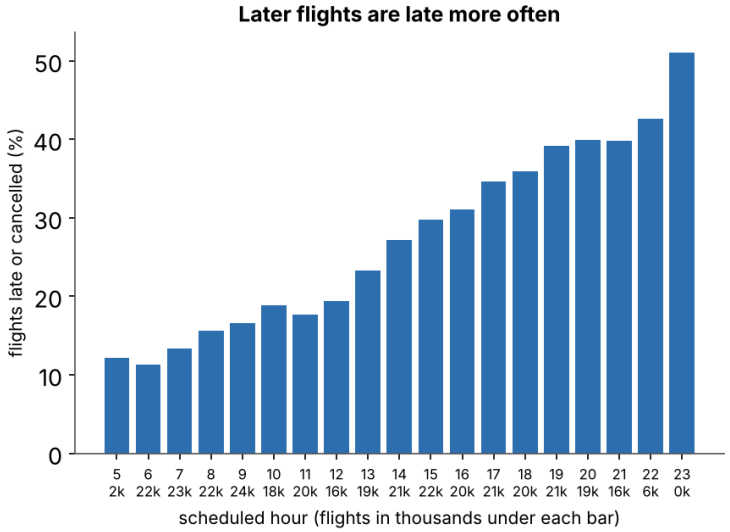
Percent of all scheduled flights that month. Orange = above 30%. [04]

- **22.9%** of the year's 435,352 flights were late (over 15 minutes) or cancelled. [04]
- In July it was **39%**, and **8.7%** of flights were cancelled. [04]
- The average client waits **13.8 minutes**; **8.4%** of departing flights are over an hour late. [04]

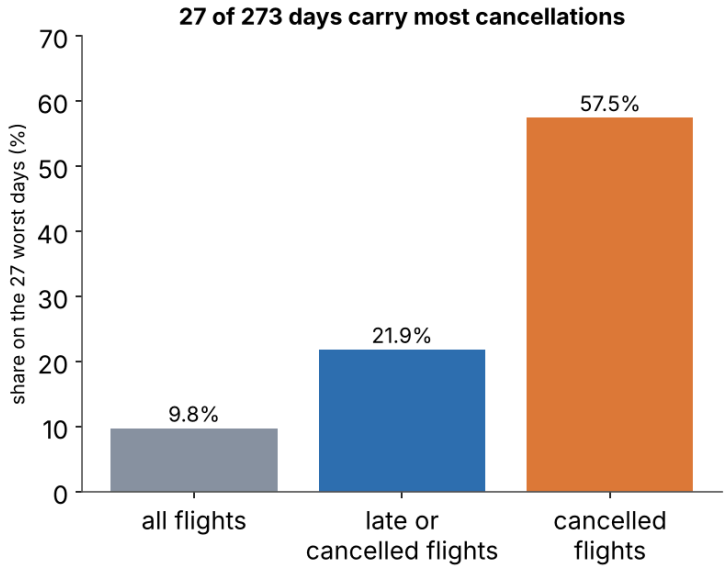
> **Today:** a client finds out at the airport. The agency cannot rebook or reassure someone it did not warn.

Source: notebook 04 (metrics M1 to M5), whole year

Three things the agency did not know, and each one is a chance to act



Delays build up through the day. 13.4% of flights at 5-8 h are late or cancelled, **39.9%** at 19-22 h (gap 26.5 points). [02]



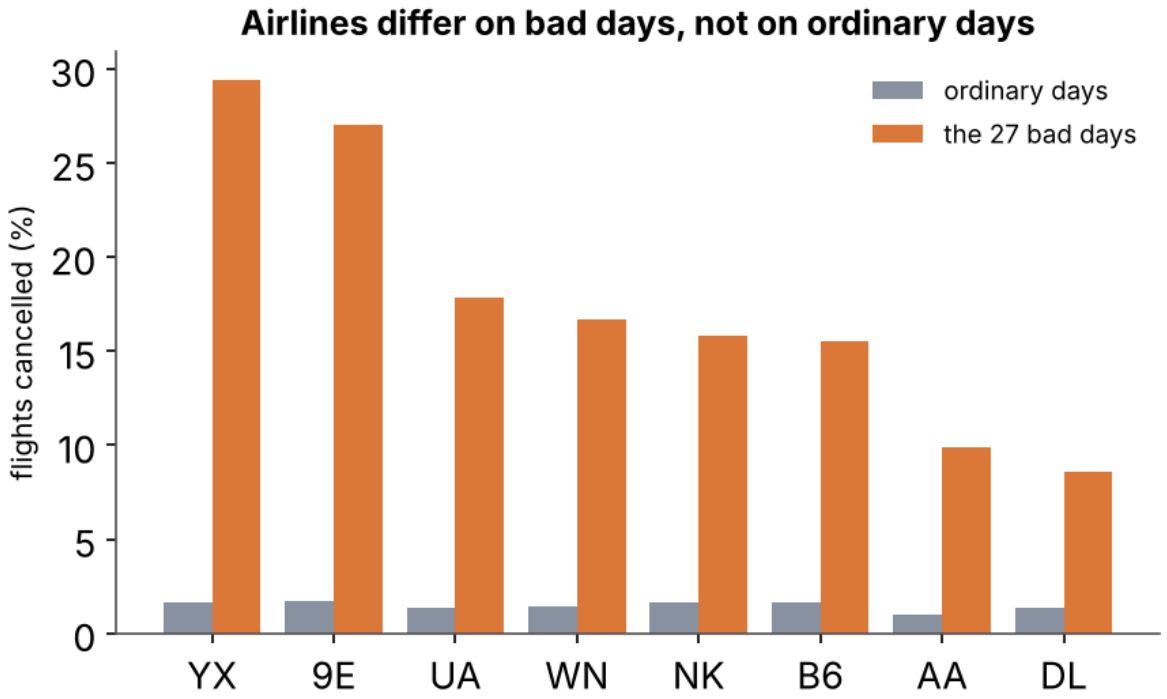
A few bad days do most of the damage. 27 days out of 273 hold **57.5%** of all cancellations. [02]

The state of the airport beats the weather. How late the flights that just left the same airport were is the strongest warning sign, 10 times stronger than weather (0.228 against 0.023). [06]

Rain adds 14.5 points of late flights, but most bad days are not rainy days. [02]

Sources: notebooks 02, 06 (training months, Jan-Sep)

The risk is concentrated, so the action can be targeted



On a bad day Republic (YX) cancels **29%** and Endeavor (9E) **27%** of flights, Delta 9%. On ordinary days all big airlines cancel 1% to 1.7%. [02]

Destinations	Late rate	Do this
Main long-haul (45 airports, 226,896 flights)	27.8%	warn earlier on afternoon and evening flights
Short-haul (57 airports, 101,287 flights)	21.2%, most cancellations (4.3%)	have rebooking ready first
Chronic delays (7 airports, 2,374 flights)	53.1%	always warn, check connections

Destination groups from k-means on delay patterns; the structure is weak (silhouette 0.322), so treat them as a guide. [03]

Sources: notebooks 02 (airlines), 03 (groups), Jan-Sep

How it works: two hours before departure the system scores every flight and the agency acts on the flagged ones

1. Two hours before
Look at what is known then: time, airline, route, weather, how late the airport and the plane already are.

→

2. Score every flight
A risk score for each flight: late or cancelled (Alert 1) and cancelled (Alert 2).

→

3. Flag above the cut-off
Cut-offs were set on July to September, before the test months.

→

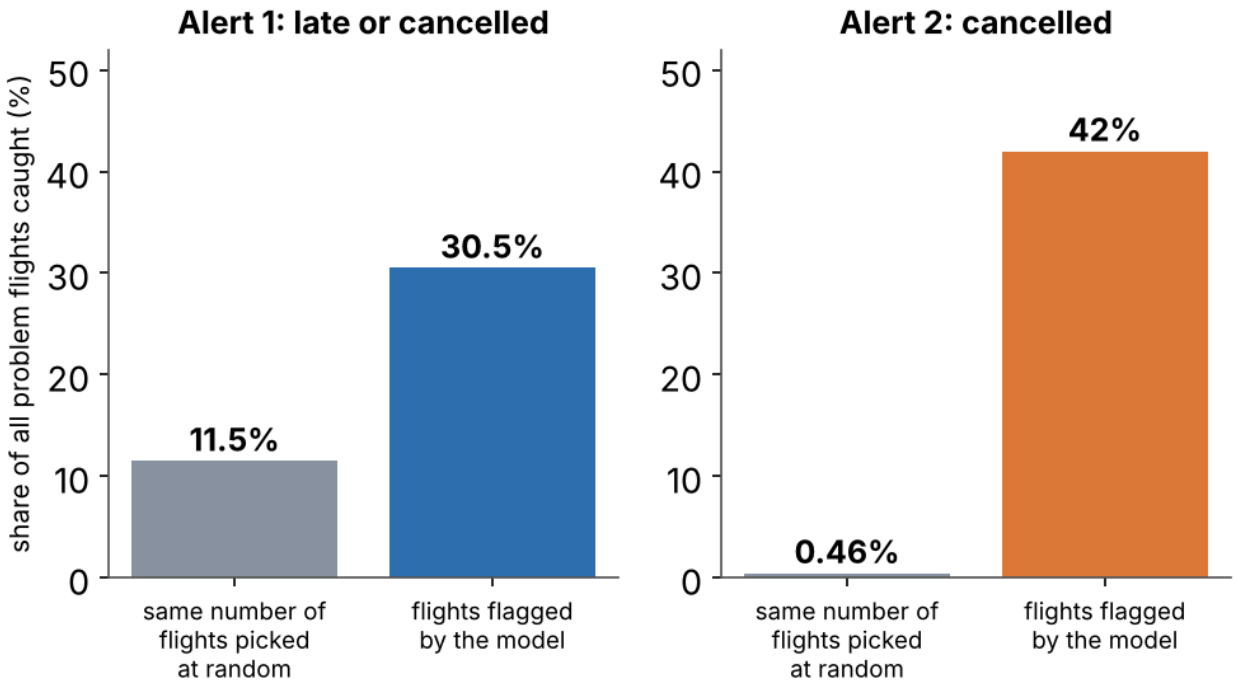
4. A person acts
Check the airline's feed, then message the client or offer a rebooking. Nothing is automatic.

No cheating: columns written after the flight (actual times, delays, the tail number of cancelled flights) are never used; four automated tests prove no information from after the prediction moment gets in. [05]

Tested honestly: trained on January to September, scored once on October to December (104,469 flights). The "do nothing" answer is the bar to beat. [06]

Sources: notebooks 05 (leak tests), 06 (protocol)

It works: the flagged flights catch far more problems than picking the same number at random



October to December 2023, flights the model never saw. Random pick = same number of flights chosen by chance. [07]

- **Alert 1** catches **30.5%** of late or cancelled flights while flagging 11.5% of flights: **2.7 times** better than random. [07]
- **Alert 2** catches **42%** of cancellations while flagging 0.46% of flights: **91 times** better than random. [07]

> Doing nothing catches **0**. The best "do nothing" score is 0.141 (alert 1) and 0.0021 (alert 2); the model scores 0.339 and 0.120. [06]

Source: notebook 07, section 7 (cards); notebook 06, section E

Alert 1: message about 130 flights a day, and three in eight of them really go wrong

130
flights flagged a day (11,974 in three months) [07]

37.4%
of flagged flights were really late or cancelled (4,481), against 14.1% for a random flight [07]

51%
of evening flights (19-23 h) that go wrong are caught [07]

- **Action:** message the client two hours before and check their connections; warn earlier on afternoon and evening flights.
- **Why it pays:** a false alarm costs one message; a missed delay leaves a client unwarned.
- **Honest limit:** mornings are hard (5.8% caught) and Newark is weaker (20.5%). [07]
- **Human check:** confirm the flight status first; the message says "possible delay".

Source: notebook 07, section 7 (card 1); 104,469 test flights, Oct-Dec

Alert 2: about five flights a day to prepare rebooking for, catching 42% of cancellations

5.3
flights flagged a day (484 in three months) [07]

94
of the 224 cancellations found (42%) [07]

1 in 5
flags is a real cancellation (19.4%); 1 in 500 for a random flight [07]

- **Action:** offer a rebooking to the client of a flagged flight before the airline cancels; strongest for short-haul (25.1% precise).
- **Use it as a ranking:** it tells the team which flights to look at first.
- **Honest limit:** 115 of the 224 cancellations fell on one day (18 December), so the result is uncertain (interval 1% to 60% recall). [07]
- **Never auto-rebook.** A flag means "high risk", not "will be cancelled".

Source: notebook 07, section 7 (card 2)

What it cannot do, said plainly, so nobody is surprised

- **Mornings:** only 5.8% of problem flights are caught in the early hours. [06]
- **Newark:** 20.5% caught for Alert 1, and none of the 39 Newark cancellations for Alert 2. [06]
- **Bad days:** the cause of 18 December is not explained by the data. Weather was ordinary. [07]
- **One year, three airports:** autumn was calmer than summer (late 14.1% against 25.7%), so scores shift with the season. [01]
- **Fairness:** the model flags Alaska, American and Endeavor less than their real risk, and JetBlue most. [06]
- **Weather is observed, not forecast.** Using only weather observed two hours earlier gave the same accuracy (change -0.0016, inside the interval). [07]

None of these breaks the case. They tell us where a person must stay in the loop.

Sources: notebooks 01, 06, 07

We will know quickly if it stops working: a weekly check with a clear stop rule

	Watch weekly	Stop and re-check when
Alert 1	share of problem flights caught (14 test weeks ranged 14.3% to 51.7%)	below **14%** two weeks running [07]
Alert 2	cancellations caught, in 4-week blocks of at least 30 cancellations	below **23%** in a bad-day block (half of the 46.7% seen in validation) [07]

> **Always:** a person confirms the flight status before messaging a client, handles tight connections first, and nothing is rebooked automatically.

These thresholds rest on 14 weeks of data and are tentative; they should be reviewed after the first quarter of use. [07]

Source: notebook 07, section 7, line 9 of each card

Built with AI, checked by hand: every number can be traced back to the code

How AI helped

- drafted code and slides, helped debug (memory crash, a tie bug, a divide-by-zero)
- helped plan the decisions and word the explanations

What was checked

- counts and the 15 metric check values asserted in code
- four leak tests; test months scored once
- all 7 notebooks re-run top to bottom: same numbers
- each slide number has a line in the README ledger

Data: nycflights23 (C. Ismay, S. P. Couch, H. Wickham, 2025, CC0). Code and notebooks: the project repository.

Source: README.md (how AI was used; ledger)

The decision for today: three yeses and we start

1. **Approve Alert 1** for all flights: about 130 messages a day, with a person confirming each one.
2. **Approve Alert 2** as a rebooking-readiness list: about 5 flights a day, never automatic.
3. **Approve the weekly check** and the stop rules, owned by one named person.

Next, to make it stronger

- a real weather forecast feed
- a timestamp for when each cancellation became known
- a bad-day trigger on top of the model

What we ask of the business

- how many messages a day the team can really send
- who confirms the flight status

Sources: notebook 07 (cards), notebook 02 (finding 2: bad days need a trigger)